Original: https://sites.google.com/view/fightladder/home

Archive limitation (user-approved): all webpage text and captions are preserved, but 23 of 24 original content images returned HTTP 403 and are replaced with linked placeholders. One original animated image is available locally. Fourteen embedded Google Drive videos remain online links. The Google Sites layout is adapted to a static single-column reading/printing layout. A third-party Elfsight widget is not preserved. The webpage publication date is not stated.

FightLadder

A Benchmark for Competitive Multi-Agent Reinforcement Learning

Wenzhe Li Zihan Ding Seth Karten Chi Jin

Princeton University

[Paper] [Code]

Abstract

Recent advances in reinforcement learning (RL) heavily rely on a variety of well-designed benchmarks, which provide environmental platforms and consistent criteria to evaluate existing and novel algorithms. Specifically, in multi-agent RL (MARL), a plethora of benchmarks based on cooperative games have spurred the development of algorithms that improve the scalability of cooperative multi-agent systems. However, for the competitive setting, a lightweight and open-sourced benchmark with challenging gaming dynamics and visual inputs has not yet been established. In this work, we present FightLadder, a real-time fighting game platform, to empower competitive MARL research. Along with the platform, we provide implementations of state-of-the-art MARL algorithms for competitive games, as well as a set of evaluation metrics to characterize the performance and exploitability of agents. We demonstrate the feasibility of this platform by training a general agent that consistently defeats 12 built-in characters in single-player mode, and expose the difficulty of training a non-exploitable agent without human knowledge and demonstrations in two-player mode. FightLadder offers meticulously crafted environments to tackle essential challenges in competitive MARL research, heralding a new era of discovery and advancement.

Supported Games
Single-Agent RL Experiments
Multi-Agent RL Experiments
RL Exploitation
Human Exploitation
Head-to-Head Evaluation
Elo Rating
Citation

Supported Games

[Archive limitation: original image 1 could not be downloaded (HTTP 403). Original image link]

Mortal Kombat (Genesis)

原文：https://sites.google.com/view/fightladder/home

存档限制（经用户批准）：所有网页文本和图片说明均已保留，但原始内容图片共 24 张，其中 23 张返回 HTTP 403，已替换为带链接的占位符。一张原始动图可在本地查看。十四个嵌入式 Google 云端硬盘视频仍以在线链接形式保留。Google 协作平台的页面布局已调整为适合阅读和打印的静态单栏布局。第三方 Elfsight 小组件未予保留。网页未注明发布日期。

FightLadder

竞争性多智能体强化学习基准

Wenzhe Li Zihan Ding Seth Karten Chi Jin

普林斯顿大学

[论文] [代码]

摘要

强化学习（RL）的近期进展在很大程度上依赖于各种精心设计的基准，这些基准提供了环境平台和一致的标准，用于评估现有算法和新算法。具体而言，在多智能体强化学习（MARL）领域，大量基于合作博弈的基准推动了相关算法的发展，提升了合作式多智能体系统的可扩展性。然而，在竞争场景下，尚未建立兼具挑战性游戏动态机制与视觉输入的轻量级开源基准。在本研究中，我们提出了实时格斗游戏平台 FightLadder，以助力竞争性多智能体强化学习研究。除该平台外，我们还提供了面向竞争博弈的前沿多智能体强化学习算法的实现，以及一套用于衡量智能体性能和可利用性的评估指标。我们训练了一个能在单人模式下稳定击败 12 个内置角色的通用智能体，从而验证了该平台的可行性；同时，我们也揭示了在双人模式下，不借助人类知识和示范来训练不可利用的智能体的难度。FightLadder 提供了精心构建的环境，以应对竞争性多智能体强化学习研究中的关键挑战，开启了探索与进步的新篇章。

支持的游戏 单智能体强化学习实验 多智能体强化学习实验 强化学习利用 人类利用 直接对战评估 埃洛评分 引用

支持的游戏

[归档限制：无法下载原始图片 1（HTTP 403）。原始图片链接]

真人快打（世嘉 Genesis）



Fatal Fury 2 (Genesis)

[Archive limitation: original image 2 could not be downloaded (HTTP 403). Original image link]

The King of Fighters '97 (Neo Geo)

[Archive limitation: original image 3 could not be downloaded (HTTP 403). Original image link]

Street Fighter II (Genesis)

[Archive limitation: original image 4 could not be downloaded (HTTP 403). Original image link]

Street Fighter III (Arcade)

Single-Agent RL Experiments

In the single-player setting, we proposed a learning scheme based on **curriculum learning**. It trains a general RL agent (*Proximal Policy Optimization*, **PPO** in our experiments) that can consistently beat built-in computer players (CPUs) across different characters. For game *Street Fighter II*, the training process of the agent over 25 epochs is shown below. Each epoch involves 10M training steps competing with opponents sampled from the curriculum scheduler in parallel.

[Archive limitation: original image 5 could not be downloaded (HTTP 403). Original image link]

The following video shows the learned single-agent PPO (left, Ryu, in grey) is capable of **passing through all 15 levels** of the *Street Fighter II* game.

[Archive limitation: embedded video not saved offline: target_16_episode_1_done_16.mp4. Watch original video online]

Multi-Agent RL Experiments

For the two-player setting (controlling both sides), we adopt five MARL algorithms for agent training, including three population-based methods and two decentralized methods:

- **League Training (e.g., AlphaStar)**
- **Policy Space Response Oracle (PSRO)**
- **Fictitious Self-Play (FSP)**
- **Independent Proximal Policy Optimization (IPPO)**
- **Two-Timescale IPPO**.



饿狼传说2（世嘉 Genesis）

[存档限制：无法下载原始图片 2（HTTP 403）。原始图片链接]

拳皇97（Neo Geo）

[归档限制：无法下载原始图片 3（HTTP 403）。原始图片链接]

街头霸王II（世嘉 Genesis）

[归档限制：无法下载原始图片 4（HTTP 403）。原始图片链接]

街头霸王III（街机）

单智能体强化学习实验

在单人游戏设置中，我们提出了一种基于**课程学习**的学习方案。该方案训练一个通用强化学习智能体（在我们的实验中采用近端策略优化，**近端策略优化**），使其能够在面对不同角色时稳定击败内置电脑玩家（电脑玩家）。下图展示了该智能体在街头霸王II游戏中经过25个训练轮次的训练过程。每个训练轮次包含1000万训练步数，智能体并行地与从课程调度器中采样的对手对战。

[存档限制：无法下载原始图片5（HTTP 403）。原始图片链接]

以下视频展示了训练后的单智能体近端策略优化智能体（左侧，身穿灰衣的隆）能够**通关全部15个关卡**，完成街头霸王II游戏。

[A存档限制：嵌入式视频未离线保存：目标_16_回合_1_完成_16.mp4。在线观看原始视频]

多智能体强化学习实验

对于双玩家设置（控制双方），我们采用五种多智能体强化学习算法进行智能体训练，包括三种基于种群的方法和两种去中心化方法：

- **联赛训练（例如 AlphaStar）**
- **策略空间响应预言机（PSRO）**
- **虚拟自博弈（FSP）**
- **独立近端策略优化（IPPO）**
- **双时间尺度独立近端策略优化**。

For three **population-based methods (league training, PSRO, and FSP)**, the dynamics of payoff matrices within each learning population during training are shown as follows. The name of each row indicates the agent information as *Character_Side_Checkpoint*. *Checkpoint=h_xM* represents a previous version of the agent saved at *x* million steps. The value indicates the win rate of the left (row) player against the right (column) player. Please refer to more details in our paper.

[Archive limitation: original image 6 could not be downloaded (HTTP 403). Original image link]

League Training

[Archive limitation: original image 7 could not be downloaded (HTTP 403). Original image link]

PSRO

[Archive limitation: original image 8 could not be downloaded (HTTP 403). Original image link]

FSP

RL Exploitation

To examine the exploitability of learned agents with the above MARL algorithms, we use a single-agent RL algorithm (PPO) to train the exploiter model against the **fixed** best agent from each multi-agent algorithm. Videos of **learned agents** (left, Ryu, in grey) versus **RL exploiters** (right, Ryu, in white) are shown below.

In general, after sufficient time of training, the exploiter can eventually beat learned agents consistently, while agents from different algorithms reveal various weaknesses.

[Archive limitation: embedded video not saved offline: exploit_ppo_left.mp4. Watch original video online]

PPO (Single-Agent Baseline) vs **PPO Exploiter**

[Archive limitation: embedded video not saved offline: exploit_ippo_left.mp4. Watch original video online]

IPPO vs **PPO Exploiter**

[Archive limitation: embedded video not saved offline: exploit_2timescale_left.mp4. Watch original video online]

Two-timescale IPPO vs **PPO Exploiter**

[Archive limitation: embedded video not saved offline: exploit_league_left.mp4. Watch original video online]

League Training vs **PPO Exploiter**

[Archive limitation: embedded video not saved offline: exploit_psro_left.mp4. Watch original video online]

PSRO vs **PPO Exploiter**

[Archive limitation: embedded video not saved offline: exploit_fsp_left.mp4. Watch original video online]

FSP vs **PPO Exploiter**

Human Exploitation

In addition to exploiting the learned agents with RL algorithms, we also attempt to exploit them with human players under the same control frequency. Videos of **learned agents** (left, Ryu, in gray) versus **humans** (right, Ryu, in white) are shown below.

In general, our human players choose a *defensive counterattack* strategy instead of a fully aggressive one, which is found effective in exploiting the learned agents across methods.

对于三种**基于种群的方法（联赛训练、策略空间响应预言机和虚拟自博弈）**，各学习种群在训练过程中的收益矩阵变化如下所示。每行的名称按角色_阵营_检查点的格式标明智能体信息。检查点=*h_xM*表示在训练至 *x* 百万步时保存的智能体历史版本。矩阵中的数值表示左侧（行）玩家对右侧（列）玩家的胜率。更多详细信息请参阅我们的论文。

[归档限制：无法下载原始图片 6（HTTP 403）。原始图片链接]

联赛训练

[归档限制：无法下载原始图片 7（HTTP 403）。原始图片链接]

PSRO

[归档限制：无法下载原始图片 8（HTTP 403）。原始图片链接]

FSP

强化学习利用

为检验使用上述多智能体强化学习算法训练得到的智能体的可利用性，我们使用单智能体强化学习算法（近端策略优化）训练利用者模型，使其对抗每种多智能体算法中**固定不变的**最佳智能体。下方展示了**已训练智能体**（左侧，灰衣隆）与**强化学习利用者**（右侧，白衣隆）的对战视频。

总体而言，经过充分训练后，利用者最终能够稳定击败已训练智能体，而不同算法训练出的智能体则暴露出各自的弱点。

[存档限制：嵌入式视频未离线保存：利用_近端策略优化_左侧.mp4。在线观看原始视频]

近端策略优化（单智能体基线）对阵**近端策略优化利用者**

[存档限制：嵌入式视频未离线保存：利用_独立近端策略优化_左侧.mp4。在线观看原始视频]

独立近端策略优化对阵**近端策略优化利用者**

[存档限制：嵌入式视频未离线保存：利用_2时间尺度_左侧.mp4。在线观看原始视频]

双时间尺度独立近端策略优化 对阵 **近端策略优化利用者**

[归档限制：嵌入式视频未离线保存：利用_联赛_左侧.mp4。在线观看原始视频]

联赛训练 对阵 **近端策略优化利用者**

[归档限制：嵌入式视频未离线保存：利用_策略空间响应预言机_左侧.mp4。在线观看原始视频]

策略空间响应预言机 对阵 **近端策略优化利用者**

[归档限制：嵌入式视频未离线保存：利用_虚拟自博弈_左侧.mp4。在线观看原始视频]

虚拟自博弈 对阵 **近端策略优化利用者**

人类利用

除了使用强化学习算法利用已训练智能体的弱点外，我们还尝试让人类玩家在相同的控制频率下利用这些智能体的弱点。下方展示了**已训练智能体**（左侧，灰衣隆）对阵**人类玩家**（右侧，白衣隆）的视频。

总体而言，我们的人类玩家选择防守反击策略，而非全面进攻策略；事实表明，这一策略能够有效利用各种方法训练出的已训练智能体的弱点。

Nevertheless, we still find that the learned agents reveal some robustness to human players (e.g., defend when a human player attacks) during human evaluations, but some simple strategies (e.g., defensive posture combined with low kicks at proper timing) could still defeat them rather consistently.

[Archive limitation: embedded video not saved offline: ppo.mp4. Watch original video online]

PPO (Single-Agent Baseline) vs **Human**

[Archive limitation: embedded video not saved offline: ippo.mp4. Watch original video online]

IPPO vs **Human**

[Archive limitation: embedded video not saved offline: 2timescale.mp4. Watch original video online]

Two-timescale IPPO vs **Human**

[Archive limitation: embedded video not saved offline: league.mp4. Watch original video online]

League Training vs **Human**

[Archive limitation: embedded video not saved offline: psro.mp4. Watch original video online]

PSRO vs **Human**

[Archive limitation: embedded video not saved offline: fsp.mp4. Watch original video online]

FSP vs **Human**

Head-to-Head Evaluation

We further evaluate the trained MARL agents with head-to-head battles among each other, as well as against CPUs (built-in game players). The recorded videos are shown below, with Ryu as characters for both sides.

[Archive limitation: original image 9 could not be downloaded (HTTP 403). Original image link]

IPPO vs **CPU**

[Archive limitation: original image 10 could not be downloaded (HTTP 403). Original image link]

Two-timescale IPPO vs **CPU**

[Archive limitation: original image 11 could not be downloaded (HTTP 403). Original image link]

League Training vs **CPU**

[Archive limitation: original image 12 could not be downloaded (HTTP 403). Original image link]

PSRO vs **CPU**

[Archive limitation: original image 13 could not be downloaded (HTTP 403). Original image link]

FSP vs **CPU**

[Archive limitation: original image 14 could not be downloaded (HTTP 403). Original image link]

PSRO vs **IPPO**

[Archive limitation: original image 15 could not be downloaded (HTTP 403). Original image link]

League Training vs **PSRO**

尽管如此，我们仍发现，在人类评估中，已训练智能体面对人类玩家时表现出一定的鲁棒性（例如，在人类玩家发起攻击时进行防御），但一些简单的策略（例如，保持防御姿态并在适当时机配合低踢）仍能相当稳定地击败它们。

[归档限制：嵌入式视频未离线保存：ppo.mp4。 在线观看原始视频]

近端策略优化（单智能体基线）对战**人类**

[归档限制：嵌入式视频未离线保存：ippo.mp4。 在线观看原始视频]

独立近端策略优化对战**人类**

[归档限制：嵌入式视频未离线保存：2timescale.mp4。 在线观看原始视频]

双时间尺度独立近端策略优化对战**人类**

[归档限制：嵌入的视频未离线保存：league.mp4。 在线观看原始视频]

联赛训练 对战 **人类**

[归档限制：嵌入的视频未离线保存：psro.mp4。 在线观看原始视频]

策略空间响应预言机 对战 **人类**

[归档限制：嵌入的视频未离线保存：fsp.mp4。 在线观看原始视频]

虚拟自博弈 对战 **人类**

直接对战评估

我们进一步通过训练后的多智能体强化学习智能体之间的直接对战，以及它们与电脑玩家（内置游戏玩家）的对战，评估这些智能体的表现。录制的视频如下所示，双方均使用隆作为角色。

[归档限制：无法下载原始图片 9（HTTP 403）。原始图片链接]

独立近端策略优化 对战 **电脑玩家**

[归档限制：无法下载原始图片 10（HTTP 403）。原始图片链接]

双时间尺度独立近端策略优化 对战 **电脑玩家**

[归档限制：无法下载原始图片 11（HTTP 403）。原始图片链接]

联赛训练 对战 **电脑**

[归档限制：无法下载原始图片 12（HTTP 403）。原始图片链接]

策略空间响应预言机 对战 **电脑**

[归档限制：无法下载原始图片 13（HTTP 403）。原始图片链接]

虚拟自博弈 对战 **电脑**

[存档限制：无法下载原始图片 14（HTTP 403）。原始图片链接]

策略空间响应预言机 对阵 **独立近端策略优化**

[存档限制：无法下载原始图片 15（HTTP 403）。原始图片链接]

联赛训练 对比 **策略空间响应预言机**

[Archive limitation: original image 16 could not be downloaded (HTTP 403). Original image link]

Two-timescale IPPO vs **PSRO**

[Archive limitation: original image 17 could not be downloaded (HTTP 403). Original image link]

FSP vs **League Training**

Elo Rating

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Elo rating is commonly used as a ranking system for multi-player games. We assess Elo ratings for trained agents in our experiments to evaluate the learning process and agent performances, over five algorithms **league training, PSRO, FSP, IPPO, and two-timescale IPPO**.

The overall Elo ratings **with a mixed population** of agents (top 5%) from five algorithms are shown on the left. The horizontal axis is the Elo score, and the vertical axis is the number of agents with certain Elo for each algorithm. League training achieves higher Elo ratings on average, especially for left-side agents. However, we also noticed that two sides of agents reveal asymmetric strengths in terms of Elos.

The Elo ratings **within each algorithm population** are shown below.

For each algorithm, the *left* figure shows the Elo rating versus the winning rate of each policy in evaluation, the *middle* figure shows the learning progress with each point indicating a policy with the corresponding Elo, and the *right* figure depicts the Elo distribution for each type of agent.

The Elo rating for the population of agents trained with the **league training** algorithm:

[Archive limitation: original image 19 could not be downloaded (HTTP 403). Original image link]

The Elo rating for the population of agents trained with **PSRO** algorithm:

[Archive limitation: original image 20 could not be downloaded (HTTP 403). Original image link]

The Elo rating for the population of agents trained with **FSP** algorithm:

[Archive limitation: original image 21 could not be downloaded (HTTP 403). Original image link]

The Elo rating for the population of agents trained with **IPPO** algorithm:

[Archive limitation: original image 22 could not be downloaded (HTTP 403). Original image link]

The Elo rating for the population of agents trained with **two-timescale IPPO** algorithm:

[Archive limitation: original image 23 could not be downloaded (HTTP 403). Original image link]

[Archive limitation: embedded video not saved offline: fsp_(left)_vs_fsp_.mp4. Watch original video online]

FSP vs **FSP**

[存档限制：无法下载原始图片 16（HTTP 403）。原始图片链接]

双时间尺度独立近端策略优化 对比 **策略空间响应预言机**

[存档限制：无法下载原始图片 17（HTTP 403）。原始图片链接]

虚拟自博弈对比**联赛训练**

埃洛评分

[存档限制：无法下载原始图片 18（HTTP 403）。原始图片链接]

埃洛评分通常用作多人游戏的排名系统。在实验中，我们评估了采用五种算法**（联赛训练、策略空间响应预言机、虚拟自博弈、独立近端策略优化和双时间尺度独立近端策略优化）**训练后的智能体的埃洛评分，以评价学习过程和智能体表现。

左图展示了来自五种算法的智能体（排名前5%）**组成混合种群时**的整体埃洛评分。横轴为Elo分数，纵轴为每种算法中具有特定Elo分数的智能体数量。联赛训练取得的平均埃洛评分更高，尤其是左侧智能体。不过，我们也注意到，双方智能体在Elo分数所反映的实力上存在不对称性。

下图展示了**各算法种群内部**的埃洛评分。

对于每种算法，左图展示了评估中各策略的埃洛评分与胜率之间的关系，中图展示了学习进展，其中每个点表示一个策略及其对应的Elo分数，右图则描绘了各类智能体的Elo分布。

使用**联赛训练**算法训练的智能体种群的埃洛评分：

[存档限制：无法下载原始图片19（HTTP 403）。原始图片链接]

使用**策略空间响应预言机**算法训练的智能体种群的埃洛评分：

[存档限制：无法下载原始图片 20（HTTP 403）。原始图片链接]

使用**虚拟自博弈**算法训练的智能体种群的埃洛评分：

[存档限制：无法下载原始图片 21（HTTP 403）。原始图片链接]

使用**独立近端策略优化**算法训练的智能体种群的埃洛评分：

[存档限制：无法下载原始图片 22（HTTP 403）。原始图片链接]

使用**双时间尺度独立近端策略优化**算法训练的智能体种群的埃洛评分：

[归档限制：无法下载原始图片 23（HTTP 403）。原始图片链接]

[存档限制：嵌入的视频未离线保存：fsp_（左侧）_对战_fsp_.mp4。在线观看原始视频]

虚拟自博弈 对阵 **虚拟自博弈**

An interesting observation from above Elo rating during entire training process is that, the agents learned on left and right sides can be asymmetric in their performances. This is further verified in our experiments with head-to-head test for agents on two sides but with the same MARL algorithm, for example, video of two FSP agents.

Citation

```
@misc{li2024fightladder,
      title={FightLadder: A Benchmark for Competitive Multi-Agent Reinforcement Learning},
      author={Wenzhe Li and Zihan Ding and Seth Karten and Chi Jin},
      year={2024},
      eprint={2406.02081},
      archivePrefix={arXiv},
      primaryClass={cs.MA}
}
```


从上述整个训练过程中的埃洛评分可以观察到一个有趣的现象：在左右两侧训练的智能体，其表现可能存在不对称性。我们在实验中，让位于两侧但采用相同多智能体强化学习算法的智能体进行直接对战，进一步验证了这一点，例如两个虚拟自博弈智能体的对战视频。

引用

```
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